YULIM (JESS) CHOI

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EDUCATION

University of Southern California, Viterbi School of Engineering — **Los Angeles, CA**
Master of Science in Applied Data Science — Aug. 2026 – May 2028

Sungkyunkwan University (SKKU) — **Seoul, Korea**
Bachelor of Library and Information Science; Bachelor of Business Administration — Feb. 2022 – Feb. 2026

- GPA: 3.86/4.00 (Magna Cum Laude)
- Thesis: "From Metadata Gaps to a Reuse-Oriented Data Lifecycle Model: A Quantitative Analysis of the National Library of Korea's Bibliographic LOD Using DCC and FAIR"
- Academic Excellence Scholarship – Top 3% (Highest Honors, Spring 2023); Top 5% (Honors, Fall 2024), College of Liberal Arts

Indiana University, Kelley School of Business — **Bloomington, IN**
Exchange Student — Aug. 2024 – May 2025

TECHNICAL SKILLS

Python | SQL | R | Microsoft Excel | Tableau

EXPERIENCE

Department of Library and Information Science, SKKU — **Seoul, Korea**
Undergraduate Research Assistant (Advisor: Prof. Inkyung Choi) — Oct. 2025 – Aug. 2026

- Built an end-to-end pipeline transforming raw ECG waveforms into an analysis-ready dataset of 35,980 observations and 13 HRV features through quality control, feature engineering, aggregation, and cross-cohort standardization.
- Built a Python-based preprocessing workflow for 56 full-text materials science XML papers, performing section-level parsing, text normalization, and sentence segmentation to generate a structured 4,622-sentence corpus for Named Entity Recognition.
- Implemented data cleaning and deduplication for the NER corpus, reducing 18,488 text examples to 3,782 unique examples (79.5% reduction) prior to Doccano annotation.
- Developed an end-to-end literature-to-annotation pipeline integrating API/XML acquisition, text cleaning, sentence-level chunking, and NER pre-annotation into Doccano-ready JSONL outputs.
- Administered the Doccano annotation environment on a university server, managing dataset import/export, user access, annotation records, and SQLite database backups.

ACADEMIC PROJECTS

Analyzing Influential Factors of Traffic Stops in San Francisco | *R*

- Built a decision tree model and binomial simulation to detect demographic/contextual factors affecting traffic stop outcomes.
- Cleaned and transformed 900K+ rows, reducing dataset size by 10% to optimize model efficiency and minimize load times.
- Simulated 10,000 traffic stop scenarios to quantify disparities and assess enforcement risks with statistical precision.

Passenger Congestion Prediction at Hyehwa Subway Station| *Python*

- Built predictive models using ML algorithms such as K-Nearest Neighbors, Linear Regression, and Random Forest to forecast daily congestion levels at Hyehwa Station.
- Consolidated fragmented ridership data into a unified dataset, integrating temporal and seasonal features.
- Visualized passenger flow patterns using Seaborn and Matplotlib to uncover peak congestion periods and seasonal variability.

LEADERSHIP ACTIVITIES

Hult Prize On Campus, SKKU – *Campus Director* — Oct. 2023 – Mar. 2024

- Spearheaded SKKU's first-ever Hult Prize competition, engaging 76 applicants and advancing 13 finalist teams to address UN Sustainable Development Goals.
- Directed end-to-end program execution, including team recruitment, applicant screening, event promotion, mentoring, and coordination with mentors, judges, professors, World Vision Korea CFO, and startup CEO.
- Oversaw final competition logistics, budgeting, and communications with SKKU administration and Hult Prize HQ for seamless program delivery; organized intercollegiate networking events with 68 participants from 3 universities.